

SSCD Vignette on GtexResampled Data

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Load SSCD

```
library(SSCD)
```

Load GtexResampled Data

```
data("GtexResampled", package = "SSCD")
```

Load the expression matrix

SSCD optimization is fast (minutes) in a multi-core environment. We demonstrate with a small subset.

```
small_mix <- Dataset_Resampled_Gtex$ex[1:100, ]
```

Get the number of compartments from the trueH matrix

```
compartment_n <- nrow(Dataset_Resampled_Gtex$trueH)
```

Run SSCD

Set the number of cores (parallel_n) available for the optimization.

```
improvedW <- SSCD::SSCD(small_mix, compartments_n = compartment_n, nrun = 200, nmf_seed = 124578,  
  mvg = 1000, parallel_n = 10)
```